Kutty Banerjee

kutty.banerjee@gmail.com (925)-209-7204 (US Citizen)

Experienced and interested in Operating Systems, Software-Hardware Interface, Performance Modeling and Parallel Computing.

Languages : (C / Java) (C++ / Objective-C / Verilog)

Apple: iOS GPU Firmware Engineer [Aug 2013 – Present]

GPU Scheduling Lead

- Implemented an improved GPU Scheduler from scratch. It is Credit based, provides Fairness across Multiple Priority Bands and allows Deadline based Preemption to maintain a minimum Guaranteed Frame Rate.
- Scheduler is Modular and can be run as a Standalone User Mode Binary on the CPU. Useful for Modeling impact of Tunables and algorithms on workloads. Similar to Linux Kernel's LinSched in this regard.
- Also allows capturing real world Scheduler Traces and playing them back in the Model to study Scheduling behavior.
- Patents filed.

Firmware Unit Test Infrastructure

- Added Unit Test support in the Firmware. Converting more and more Firmware code into Isolated Modules for Unit Testing everyday. Also added support for Unit Randoms to flush out Concurrency bugs.

Power Model

- Maintained and augmented Firmware PID Controller Logic for iPad Air2, iPhone 6/6+. Used for Predicting Runtime power consumption based on Perf Counter Utilization which feeds back into the GPU Thermal Controller.

Kernel / Firmware

- Contributed to Kernel Memory Management, OpenCL Driver.
- Added Data Flow Logic in OpenCL to elide Cache Flushes between back to back OpenCL kernels that do not have WAW / RAW dependencies.
- Helped reduce Interrupt servicing time latency by reducing icache and dcache misses in the handler.
- Did tons of Performance triage and Performance fixes of the Firmware

TraceAnalyzer

- Implemented a Java Swing + JFreeChart based tool that takes in Firmware and Hardware traces and runs Timeseries Queries off them to display Latency Maps, Causal Relationships between Interrupts and Processing and Scheduling .
- Useful for Performance bottleneck hunting in Embedded Operating Systems and Firmware.

Cyclops - Cycle Accurate Simulator for Performance and Power Modeling (Side Project)

- Implemented a Verilog like API in Java for modeling Chip Architectures. Simulator can be run in Untimed mode or in the Cycle Timed mode. The latter is useful for Perf and Power Analysis.
- In Cycle Timed Mode, Verilog like "always blocks" are Statically Scheduled by parsing Intra Module Signal Dependencies and Clock delays. This avoids Event Queueing overheads typical in common Cycle Accurate Simulators.
- Dumps out Signal changes in VCD file format for easy viewing in GTKWave and Impulse Waveform Viewers.

Intel: GPU Platform Architect [May 2013 – Aug 2013]

- Drive Low Power design of Compute features on next-gen Intel GPUs. Use models to predict Perf and Power.

NVIDIA: Software Architect [May 2005 – May 2013]

Denver CPU

Worked in the MTS Team that maintains a JIT to translate ARM instructions into a native VLIW ISA and extract better ILP via Dynamic Code Optimization. Implemented the following features

- Branch assert instruction to remove cold branches and improve instruction scheduling
- Lazy Branch Assert to avoid removing asserting branches until they are frequently taken in small windows of time.
- Branch Reprofiling to detect dynamic changes in branch weights of a running program and trigger recompilation.
- Elide Memory fence instructions. This allowed more aggressive instruction scheduling. Transaction rollback is done if a concurrent commit collides with the speculative lines. Helps JVM workloads.

Also implemented patches for Tegra Linux kernel to optimize for Denver. Changes include

- Large Page support for ARMv7. Used a BSD like Lazy Page Coalescing approach to collapse adjacent small pages into large page
- Added a new Thread Local Memory type with support in madvise and mmap system calls. Memory Descriptors marked Thread Local are treated separately in the Kernel MM. TLB invalidates, IPI broadcasts and cache snooping for these ranges can be elided on processors where doing so helps performance such as for Denver.

Project Orion - Cache Coherent CPU/GPU SoC

- Prototyped ARM64 Linux kernel changes to implement Uniform Address Space across CPU and GPUs using Shadow Page Tables.
- Worked with Hardware teams to support Page Fault and Fault aggregation.
- Project was cancelled, though a lot of the above work was carried forward to the subsequent UVM implementation.

Parallel Computing Framework (June 2011 ~ Sep 2011 Sabbatical)

Khronos - Software Implementation of OoO Processor (www.pseudoparallel.com/khronos)

- Its a stack based Virtual Machine with its own ISA. Instructions are similar to OpenCL kernels and Operands are Memory Tiles
- Instruction Parallelism is obtained by doing Data Flow Analysis of Tile Operands. The resultant Anti Dependency Graph is walked in parallel using Multiple CPU Threads.
- Supports Branches vis Predication and has first class instructions for loops. Also has a special Parallel Reduction Instruction. Supports Software Pipelining for Loop Optimizations.
- Added a Data Aware Scheduler to factor in memory traffic induced by migrating work to coprocessors.
- Added Register Renaming to break WAR stalls for Operands where possible.

Concurrency and Parallel Programming Tools / Libraries

- Mailbox based Thread Pool Implementation. Uses dedicated mail boxes each with separate locks to minimize lock contention. www.pseudoparallel.com/task-parallel-library/
- Transactional and Fence free Algorithms www.pseudoparallel.com/concurrency-utilities-c/
- Implemented a Reader lock where reader does not use an atomic instruction or fence to acquire the reader lock.
- Task Parallel Library with special smarts around Lock Contention minimizing for Dynamic Parallelism www.pseudoparallel.com/task-graph/

GPU Test Infrastructure Lead

- Lead a Cross-Team effort to unify verification efforts of various Hardware and Software/Driver teams
- Added “Probe Based Verification” to Policy-Manager, which is the layer responsible for introducing faults/events and boundary conditions that allow testing hard-to-hit conditions deterministically from an external script.
- Probes are hooks in Drivers, which can be programmed at runtime using a script to wait for asynchronous events to occur. Allows an external script to force certain hard-race conditions to occur and ensure Driver works correctly.
- Power Management Transition Module was the biggest consumer of this Verification Feature.
- Coordinated with Hardware and Software teams working on PCI-E, Display, Cache/TLB, Clocks, Core-Graphics and Video to beef up verification using the above Framework.
- Merged SW use cases with RTL HW unit Test Bench to get more directed coverage

Fermi Generation GPU (DX11 compatible) Kernel Lead

- As the Kernel Driver lead, **lead a team of 25 engineers in the Resource Manager Team** and oversaw development across Simulation, RTL, Emulation, Bringup and Product Release.

- Role included Task Assignment, Initial Triage, Interaction with Hardware and other Software teams, mentoring new Engineers and Code Review. Responsible for overall progress of the Kernel team for Fermi.
- Was the first engineer to start work on the Kernel/Firmware for Fermi. Wrote and helped large parts of the kernel code including Thread Scheduling, Interrupt Handling, Memory Management, Page Table Setup, Context Switch Support using Firmware, Cache mgmt., **TLB invalidate batching**.
- Augmented Functional Simulator's Cache and Memory modeling pipeline.

Fermi Command Buffer Scheduling Lead

- Lead the Fermi Command Buffer Scheduling unit in the Driver. Fermi added Hardware support for DX11 Per Node Command Buffers. Replaced the old Time Slice Based Scheduler with Per Node Queues.
- Added support to hold off Buffer submits (throttle) to enable Engine Power Gating
- Did plenty of code refactoring and wrote lots of tests. Scheduling module had around 90% branch Code Coverage!!

Resource Manager Graphics Driver

- Worked on various parts of the kernel driver, fixed bugs, did triaging.
- Implemented a **Non Stalling Lock Free Interrupt Tree** for routing DXG kernel notifiers without delay

Education

MS Computer Science, Worcester Polytechnic Institute, MA [2003 Aug – 2005 May]

- Thesis: Remote Execution of 3D Graphics for Mobile Systems.
- Implemented a Memory and Power Model to Predict Consumption for Graphical Workloads running on Mobile Devices.
- Recipient of WPI Institute Fellowship Award for outstanding research work. Publications in Journals and Conferences

BE Computer Science, University of Mumbai [1999 July – 2003 May]

Publications

- A Middleware Architecture for Mobile 3D Graphics
- Emmanuel Agu, Kutty Banerjee, Shirish Nilekar, Oleg Rekutin, Diane Kramer.
- International Journal of Parallel, Emergent and Distributed Systems Volume 21, Issue 3, 2006
-
- Estimating Mobile Memory Requirements and Rendering Time for Remote Execution of the Graphics Pipeline
- Kutty Banerjee, Fan Wu, Emmanuel Agu
- Proceedings of Eurographics 2005
-
- Remote Execution for 3D Graphics on Mobile Devices
- Kutty Banerjee, Emmanuel Agu
- International Conference on Wireless Networks, Communications and Mobile Computing, 2005
-
- PowerSpy: Fine-Grained Software Energy Profiling for Mobile Devices
- Kutty Banerjee, Emmanuel Agu
- International Conference on Wireless Networks, Communications and Mobile Computing, 2005
-
- A Middleware Architecture for Mobile 3D Graphics
- Emmanuel Agu, Kutty Banerjee, Shirish Nilekar, Oleg Rekutin, Diane Kramer
- Third International Workshop on Mobile Distributed Computing, 2006